



Number Properties — Elementary Algebra Week 1

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Learning Objectives

- Use the number line to add and subtract positive and negative numbers
- Apply the order of operations (PEMDAS) to simplify expressions
- Identify and classify numbers within the real number sets
- Find the absolute value of integers and rational numbers
- Arrange rational numbers in order and convert fractions to decimals

Simplify each expression completely. Show all steps and circle your final answer.

1. Use the number line to simplify:

$$(-7) + 4$$

Answer: _____

2. Use the number line to simplify:

$$(-5) - (-8)$$

Answer: _____

3. Simplify using order of operations (PEMDAS):

$$3 + 4 \times 2$$

Answer: _____

4. Simplify using PEMDAS:

$$(6 + 2)^2 \div 4 - 1$$

Answer: _____

5. Simplify using PEMDAS:

$$5 \times 3^2 - (4 + 6) \div 2$$

Answer: _____

6. Simplify using PEMDAS:

$$18 \div (2 + 1) \times 3 - 5$$

Answer: _____

7. Classify -7. Write ALL number sets it belongs to (Natural, Whole, Integer, Rational, Real):

$$-7$$

Answer: _____



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8. Classify 0. Write ALL number sets it belongs to:

0

Answer: _____

9. Classify this number. Write ALL sets it belongs to:

$\frac{3}{4}$

Answer: _____

10. Evaluate the absolute value:

$|-15|$

Answer: _____

11. Evaluate:

$-|-9| + |4|$

Answer: _____

12. Evaluate:

$|3 - 10|$

Answer: _____

13. Arrange from least to greatest:

$-\frac{1}{2}, 0.6, -0.75, \frac{1}{4}$

Answer: _____

14. Place $<$, $>$, or $=$ between the two numbers:

$-\frac{3}{4}$ quad $\underline{\hspace{1cm}}$ quad -0.8

Answer: _____

15. Convert the fraction to a decimal:

$\frac{5}{8}$

Answer: _____

16. Convert the fraction to a decimal:

$\frac{7}{9}$

Answer: _____



17. Convert the fraction to a decimal:

$$\frac{11}{4}$$

Answer: _____

18. Simplify (combines PEMDAS + absolute value):

$$2 \times |-6| - 3^2 + 1$$

Answer: _____

19. True or False: Every integer is a rational number. Explain.

True or False? $\mathbb{Z} \subseteq \mathbb{Q}$

Answer: _____

20. A submarine is at -340 ft. It rises 185 ft, then descends 60 ft. What is its final depth?

$$-340 + 185 - 60$$

Answer: _____





Topics span the full Week 1 lesson: number line, PEMDAS, number sets, absolute value, ordering rationals, and fraction-to-decimal conversion.

Solutions

1. Use the number line to simplify:

$$(-7) + 4$$

→ Start at -7, move 4 units right.

→ Land on -3.

Answer: -3

2. Use the number line to simplify:

$$(-5) - (-8)$$

→ Keep-Change-Change: $(-5) + 8$

→ Move 8 units right from -5 → land on 3.

Answer: 3

3. Simplify using order of operations (PEMDAS):

$$3 + 4 \times 2$$

→ Multiply first: $4 \times 2 = 8$

→ Then add: $3 + 8 = 11$

Answer: 11

4. Simplify using PEMDAS:

$$(6 + 2)^2 \div 4 - 1$$

→ Parentheses: $6+2=8$

→ Exponent: $8^2=64$

→ Divide: $64 \div 4=16$

→ Subtract: $16-1=15$

Answer: 15

5. Simplify using PEMDAS:

$$5 \times 3^2 - (4 + 6) \div 2$$

→ Parentheses: $4+6=10$

→ Exponent: $3^2=9$

→ Multiply: $5 \times 9=45$

→ Divide: $10 \div 2=5$

→ Subtract: $45-5=40$

Answer: 40



6. Simplify using PEMDAS:

$$18 \div (2 + 1) \times 3 - 5$$

→ Parentheses: $2+1=3$

→ Left to right: $18 \div 3 = 6$

→ Multiply: $6 \times 3 = 18$

→ Subtract: $18 - 5 = 13$

Answer: 13

7. Classify -7. Write ALL number sets it belongs to (Natural, Whole, Integer, Rational, Real):

-7

→ -7 is negative → not Natural, not Whole.

→ -7 is a whole-number value → Integer.

→ All integers are Rational and Real.

Answer: Integer, Rational, Real

8. Classify 0. Write ALL number sets it belongs to:

0

→ 0 is not a counting number → not Natural.

→ 0 is non-negative whole → Whole.

→ $\text{Whole} \subset \text{Integer} \subset \text{Rational} \subset \text{Real}$.

Answer: Whole, Integer, Rational, Real

9. Classify this number. Write ALL sets it belongs to:

$\frac{3}{4}$

→ $3/4$ is a ratio of two integers → Rational.

→ Not a whole number or integer.

→ All rationals are Real.

Answer: Rational, Real

10. Evaluate the absolute value:

$|-15|$

→ Absolute value = distance from zero.

→ $|-15| = 15$

Answer: 15

11. Evaluate:

$-|-9| + |4|$

→ $|-9| = 9$, so $-|-9| = -9$

→ $|4| = 4$

→ $-9 + 4 = -5$

Answer: -5



12. Evaluate:

$$|3 - 10|$$

→ Simplify inside: $3 - 10 = -7$

$$\rightarrow |-7| = 7$$

Answer: 7

13. Arrange from least to greatest:

$$-\frac{1}{2}, \quad 0.6, \quad -0.75, \quad \frac{1}{4}$$

→ Convert to decimals: $-0.5, 0.6, -0.75, 0.25$

→ Order: $-0.75 < -0.5 < 0.25 < 0.6$

Answer: $-0.75 < -\frac{1}{2} < \frac{1}{4} < 0.6$

14. Place $<$, $>$, or $=$ between the two numbers:

$$-\frac{3}{4} \quad \text{quad} \quad \underline{\hspace{1cm}} \quad \text{quad} \quad -0.8$$

$$\rightarrow -\frac{3}{4} = -0.75$$

→ Compare: -0.75 vs -0.8

→ -0.75 is further right on the number line $\rightarrow -\frac{3}{4} > -0.8$

Answer: $-\frac{3}{4} > -0.8$

15. Convert the fraction to a decimal:

$$\frac{5}{8}$$

$$\rightarrow \text{Divide: } 5 \div 8 = 0.625$$

Answer: 0.625

16. Convert the fraction to a decimal:

$$\frac{7}{9}$$

$$\rightarrow \text{Divide: } 7 \div 9 = 0.777... = 0.7 \text{ repeating}$$

Answer: $0.\overline{7}$

17. Convert the fraction to a decimal:

$$\frac{11}{4}$$

$$\rightarrow 11 \div 4 = 2 \text{ remainder } 3$$

$$\rightarrow 2 + \frac{3}{4} = 2.75$$

Answer: 2.75



18. Simplify (combines PEMDAS + absolute value):

$$2 \times |-6| - 3^2 + 1$$

$$\rightarrow |-6| = 6$$

$$\rightarrow 3^2 = 9$$

$$\rightarrow 2 \times 6 = 12$$

$$\rightarrow 12 - 9 + 1 = 4$$

Answer: 4

19. True or False: Every integer is a rational number. Explain.

True or False? $\mathbb{Z} \subseteq \mathbb{Q}$

$\rightarrow$ Any integer n can be written as $n/1$.

$\rightarrow$ A ratio of two integers is rational by definition.

$\rightarrow$ So every integer is also rational. TRUE.

Answer: TRUE

20. A submarine is at -340 ft. It rises 185 ft, then descends 60 ft. What is its final depth?

$$-340 + 185 - 60$$

$$\rightarrow -340 + 185 = -155$$

$$\rightarrow -155 - 60 = -215$$

$\rightarrow$ Final depth: -215 ft

Answer: -215 ft

